Curvature-Aligned Probing for Local Loss-Landscape Stabilization

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Abstract

Local loss-landscape stabilization under sample growth is typically measured either pointwise or through isotropic averaging in the full parameter space. Despite practical value, both choices probe directions that contribute little to the dominant local deformation of strongly anisotropic neural landscapes. We recast stabilization as an observational problem and introduce a unified family of criteria parameterized by an aggregation order and a probing distribution; within this family we propose a curvature-aligned criterion $\Delta_2^{(D)}$ that probes the loss increment field in the top-D eigenspace of the empirical Hessian near a trained solution. Solely from a local quadratic model, we prove that $\Delta_2^{(D)}$ preserves the $\mathcal{O}(k^{-2})$ mean-squared rate of the full-space criterion while replacing ambient-dimension curvature dependence with dependence on the subspace dimension D; a corollary gives a closed-form spectral expression and a proposition identifies the top-D eigenspace as extremal within the eigenspace-aligned family. We also derive scalable estimators based on Hessian–vector products, subspace Monte Carlo, and a closed-form Gaussian-moment proxy. On a decoder-only transformer, a curvature-aligned probe occupying a tiny fraction of parameter space already reproduces the full-space mean-squared signal to within numerical noise throughout the validated local regime, and the closed-form estimator is orders of magnitude faster than direct Monte Carlo after subspace construction.

1 Introduction

Local loss geometry is often summarized through quantities such as sharpness, curvature, or Hessian spectra, typically for a fixed trained model. Our setting is different: we ask how the empirical objective deforms locally as the training set grows. In this regime, the issue is not only which functional of the loss increment should be aggregated, but also which perturbation directions should be used to observe it. Conventional wisdom holds that local criteria should probe parameter space isotropically to avoid bias, or collapse to a single point for tractability. We argue that neither choice is necessary, and view local stabilization under sample growth as an observational problem in which the probing law is an explicit design variable (Figure 1). Concretely, this lets us ask a quantitative question that the pointwise and isotropic criteria cannot resolve: *how much of one-sample landscape deformation is concentrated in the dominant curvature modes?*

This issue is especially relevant in deep networks, where local geometry is strongly anisotropic. Empirical Hessian spectra typically concentrate much of their mass in a relatively small number of dominant directions, while much of parameter space remains weakly curved or nearly flat [Li et al., 2018a, Sagun et al., 2017, Ghorbani et al., 2019, Papyan, 2019, Xu et al., 2024]. Related work also suggests that optimization often occupies low-dimensional effective subspaces and that overparameterized models contain substantial geometric redundancy through flat or symmetry-induced directions [Gur-Ari et al., 2019, Li et al., 2018b, Simsek et al., 2021, Draxler et al., 2018,

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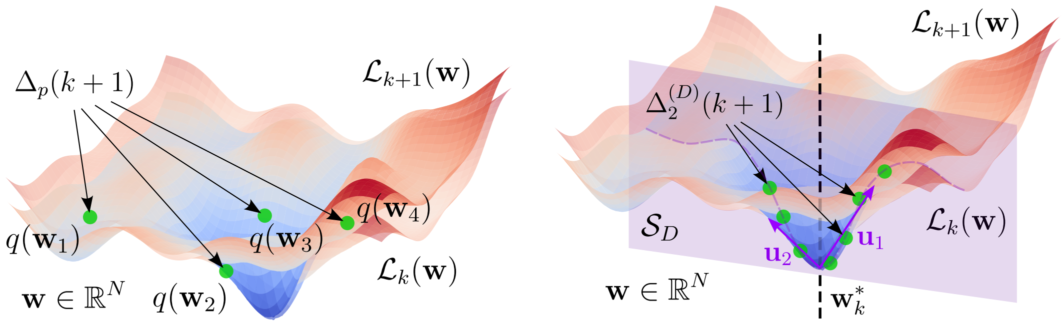

(a) Unified stabilization criterion Δ_p with probing distribution $q(\mathbf{w})$

(b) Curvature-aware subspace criterion $\Delta_2^{(D)}$

Figure 1: **Local stabilization as an observational problem.** Existing local criteria differ not only in aggregation order, but also in how they probe the increment field. Our criterion $\Delta_2^{(D)}$ restricts probing to the principal Hessian subspace spanned by the top-D curvature directions.

Garipov et al., 2018]. These observations suggest that *how* one probes local deformation should be part of the definition of stabilization, not merely part of the estimator.

We study stabilization under one-sample growth. The object of interest is the increment field $\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})$ near a trained solution $\mathbf{w}_k^*$. This places our work next to, but distinct from, classical sample-growth viewpoints. Statistical learning theory and algorithmic stability study convergence of empirical quantities, predictors, or generalization error under changes in the training set [Vapnik, 1998, Shalev-Shwartz and Ben-David, 2014, Bousquet and Elisseeff, 2002, Hardt et al., 2016, Bousquet et al., 2020]. Influence-function and infinitesimal-jackknife methods go one step closer by linking data perturbations to local first- and second-order structure [Koh and Liang, 2017, Giordano et al., 2019, Koh et al., 2019, Basu et al., 2020]. Our target is different: we study the local deformation of the *empirical objective itself*.

We formalize this viewpoint through a unified family of local stabilization criteria parameterized by an aggregation order and a probing distribution; see Eq. (1). This places previously studied one-point and isotropic mean-squared criteria into a common framework and makes the probing law part of the observable. We then propose a curvature-aware specialization, the subspace mean-squared criterion $\Delta_2^{(D)}$ in Eq. (4), which restricts probing to the top-D eigenspace of the empirical Hessian at $\mathbf{w}_k^*$.

Our main result shows that this geometric restriction does not incur a rate penalty under a local quadratic model. Theorem 2 proves that $\Delta_2^{(D)}$ preserves the $\mathcal{O}(k^{-2})$ mean-squared decay of the full-space criterion, while replacing ambient-dimension curvature dependence by dependence on the probing dimension D. In a stable-principal-directions regime, Appendix A further gives a spectral closed form and an extremality result for the top-D choice within the eigenspace-aligned family.

We also derive scalable estimators based on Hessian–vector products, subspace Monte Carlo, and a closed-form Gaussian-moment proxy. Empirically, we study how the proposed criteria decay with sample size, when the subspace criterion preserves the full-space mean-squared signal, for which perturbation scales the quadratic proxy is accurate, and how the three estimators trade fidelity against efficiency.

Contributions. Our contributions are four-fold:

- We recast local loss-landscape stabilization under sample growth as an observational problem and introduce a unified family of criteria parameterized by an aggregation order and a probing distribution.
- We propose a curvature-aware subspace criterion $\Delta_2^{(D)}$ that probes the increment field in the top-D eigenspace of the empirical Hessian near a trained solution.
- Under a local quadratic model, we prove that reducing the probe from $\mathbb{R}^N$ to a D-dimensional curvature-aligned subspace preserves the $\mathcal{O}(k^{-2})$ mean-squared rate and replaces an N-dependent curvature constant by a D-dependent one (Theorem 2).

Table 1: Positioning relative to adjacent directions. ✓ = central focus; ✗ = not a primary focus.

Direction	Data perturbation	Local objective	Geometric probe
Statistical learning theory 11, 12	✓	✗	✗
Algorithmic stability 13, 14, 20, 15	✓	✗	✗
Influence / infinitesimal jack-knife 16, 17, 18, 19	✓	✗	✗
Sharpness / SAM 21, 22, 23, 24, 25	✗	✗	✓
Hessian spectrum / subspace analysis 1, 2, 3, 4, 5, 6, 7, 26, 27	✗	✗	✓
Prior increment criteria (Δ_1, Δ_2)	✓	✓	isotropic only
This work	✓	✓	✓

- We develop scalable estimators for the criterion and show empirically that the quadratic proxy is sufficient in its validated local regime and orders of magnitude faster than direct Monte Carlo once the subspace has been constructed.

2 Related Work

Loss geometry and anisotropic local structure. A large literature studies neural loss landscapes through visualization, Hessian spectra, sharpness, and related curvature diagnostics. A recurring empirical finding is strong anisotropy: a relatively small number of eigendirections often account for much of the local second-order structure, while large parts of parameter space remain weakly curved or nearly flat [Li et al., 2018a, Sagun et al., 2017, Ghorbani et al., 2019, Papyan, 2019, Xu et al., 2024]. Related work further suggests that optimization can concentrate in low-dimensional subspaces and that overparameterized models exhibit geometric redundancy through symmetry or flat directions [Gur-Ari et al., 2019, Li et al., 2018b, Simsek et al., 2021, Draxler et al., 2018, Garipov et al., 2018]. These works motivate geometry-aware probing, but they do not study sample-growth deformation of the empirical objective.

Data perturbation, stability, and influence. Classical statistical learning theory and algorithmic stability study how empirical risks, predictors, and generalization error behave as the training sample changes [Vapnik, 1998, Shalev-Shwartz and Ben-David, 2014, Bousquet and Elisseeff, 2002, Hardt et al., 2016, Feldman and Vondrák, 2019, Bousquet et al., 2020]. A closer neighboring literature analyzes infinitesimal or finite data perturbations through influence functions and infinitesimal-jackknife approximations, linking changes in the training set to local gradient and Hessian information [Koh and Liang, 2017, Giordano et al., 2019, Koh et al., 2019, Basu et al., 2020]. These are the closest conceptual precursors to our sample-growth viewpoint. The key difference is the observable: prior work typically measures the sensitivity of fitted parameters or predictions, whereas we measure the local deformation of the empirical objective itself through the increment field $\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})$.

Sharpness, dominant eigenspaces, and second-order methods. Related work also studies flatness, sharpness, sharpness-aware optimization, and dominant Hessian eigenspaces [Keskar et al., 2017, Dinh et al., 2017, Foret et al., 2021, Dauphin et al., 2024, Luo et al., 2024, Singh et al., 2021, Wu et al., 2020]. These works ask which curvature directions matter for optimization or robustness on a fixed objective. Our question is different: given anisotropic local geometry, which probing law should be used to observe *dataset-induced* deformation? On the computational side, our estimators rely on standard scalable second-order primitives—Hessian–vector products and iterative eigensolvers—rather than explicit Hessian construction [Pearlmutter, 1994, Yao et al., 2020]. Thus, our contribution is not a new second-order tool, but a geometry-aware observable for sample-growth stabilization.

3 Method

Notation. Let $\mathfrak{D}_m = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^m$ be a training sample of size m, and let $\mathcal{L}_m(\mathbf{w}) = \frac{1}{m}\sum_{i=1}^m \ell\big(f_{\mathbf{w}}(\mathbf{x}_i), \mathbf{y}_i\big) = \frac{1}{m}\sum_{i=1}^m \ell_i(\mathbf{w})$ be the empirical risk of a parametric model, i.e., a neural network $f_{\mathbf{w}}$, with minimizer $\mathbf{w}_m^* \in \arg\min_{\mathbf{w}\in\mathbb{R}^N} \mathcal{L}_m(\mathbf{w})$. Throughout, subscripts denote per-sample quantities ($\mathbf{g}_i := \nabla \ell_i$, $\mathbf{H}_i := \nabla^2 \ell_i$), while parenthesized superscripts denote empirical averages ($\mathbf{g}^{(m)} := \nabla \mathcal{L}_m = \frac{1}{m}\sum_{i=1}^m \mathbf{g}_i$, $\mathbf{H}^{(m)} := \nabla^2 \mathcal{L}_m$). Our object of interest is the one-sample increment field $\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})$, which measures how the empirical landscape changes when one example is added. Since this increment is a scalar field over parameter space, any local notion of stabilization depends both on what is aggregated and on how the field is probed.

Unified family of stabilization criteria. For $p \geq 1$, we define a general criterion

$$\Delta_p(k+1) = \int_{\mathbb{R}^N} \left|\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})\right|^p q(\mathbf{w})\, d\mathbf{w}, \tag{1}$$

where q is a probing distribution concentrated near $\mathbf{w}_k^*$. This definition makes the probing law part of the criterion itself rather than part of the estimator. Previously studied criteria arise as special cases:

$$\Delta_1(k+1) = \left|\mathcal{L}_{k+1}(\mathbf{w}_k^*) - \mathcal{L}_k(\mathbf{w}_k^*)\right|, \qquad q = \delta_{\mathbf{w}_k^*}, \tag{2}$$

$$\Delta_2(k+1) = \int_{\mathbb{R}^N} \left(\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})\right)^2 q(\mathbf{w})\, d\mathbf{w}, \qquad q = \mathcal{N}(\mathbf{w}_k^*, \sigma^2 \mathbf{I}_N). \tag{3}$$

Curvature-aligned subspace probe. In strongly anisotropic models, isotropic perturbations spread mass across many directions that contribute little to the dominant local second-order structure [Sagun et al., 2017, Ghorbani et al., 2019, Papyan, 2019, Gur-Ari et al., 2019]. We therefore align the probe with the leading curvature directions at $\mathbf{w}_k^*$. Let $\mathbf{H}^{(k)}(\mathbf{w}_k^*) = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\lambda_1^{(k)} \geq \cdots \geq \lambda_N^{(k)}$, and let $\mathbf{U}_D = [\mathbf{u}_1, \ldots, \mathbf{u}_D]$ denote the top-D eigenvectors, with principal curvature subspace $\mathcal{S}_D = \mathrm{Im}(\mathbf{U}_D)$. We then define the *subspace mean-squared criterion*

$$\Delta_2^{(D)}(k+1) = \int_{\mathbf{w}_k^* + \mathcal{S}_D} \left(\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})\right)^2 q(\mathbf{w})\, d\mathbf{w}, \tag{4}$$

with the parameterization $\mathbf{w} = \mathbf{w}_k^* + \mathbf{U}_D \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$. The distinction between Δ_2 and $\Delta_2^{(D)}$ is therefore not merely dimensional: Δ_2 probes the increment isotropically in $\mathbb{R}^N$, whereas $\Delta_2^{(D)}$ probes it through the dominant local curvature modes of the empirical landscape.

4 Theoretical analysis

Our central theoretical finding is that geometric compression is free in rate: under the local quadratic model in Eqs. (6)–(8), the subspace criterion $\Delta_2^{(D)}$ preserves the $\mathcal{O}(k^{-2})$ decay of the full-space criterion while replacing ambient-dimension curvature dependence by dependence on the probing dimension D (Theorem 2). Within the eigenspace-aligned family, the top-D choice is moreover extremal among all D-dimensional eigenspace-aligned probes (Proposition 3).

Exact increment and local quadratic model. Adding one training example to $\mathfrak{D}_k$ gives the identity

$$\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w}) = \frac{1}{k+1}\Big(\ell_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w})\Big). \tag{5}$$

This factor $(k+1)^{-1}$ is the source of the $\mathcal{O}(k^{-2})$ scaling for squared criteria. Fix $\mathbf{w}_0 \in \mathbb{R}^N$ and assume that $\mathcal{L}_k$ and $\mathcal{L}_{k+1}$ are twice continuously differentiable near $\mathbf{w}_0$. Their second-order Taylor expansions give

$$\mathcal{L}_m(\mathbf{w}) \approx \mathcal{L}_m(\mathbf{w}_0) + \mathbf{g}^{(m)}(\mathbf{w}_0)^\top(\mathbf{w} - \mathbf{w}_0) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_0)^\top \mathbf{H}^{(m)}(\mathbf{w}_0)(\mathbf{w} - \mathbf{w}_0), \tag{6}$$

where $\mathbf{g}^{(m)} = \nabla \mathcal{L}_m$ and $\mathbf{H}^{(m)} = \nabla^2 \mathcal{L}_m$. Subtracting the two expansions yields a quadratic model for the increment field:

$$\begin{aligned} \mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w}) \approx \mathcal{L}_{k+1}(\mathbf{w}_0) - \mathcal{L}_k(\mathbf{w}_0) &+ \left(\mathbf{g}^{(k+1)}(\mathbf{w}_0) - \mathbf{g}^{(k)}(\mathbf{w}_0)\right)^\top(\mathbf{w} - \mathbf{w}_0) \\ &+ \frac{1}{2}(\mathbf{w} - \mathbf{w}_0)^\top\left(\mathbf{H}^{(k+1)}(\mathbf{w}_0) - \mathbf{H}^{(k)}(\mathbf{w}_0)\right)(\mathbf{w} - \mathbf{w}_0). \end{aligned} \tag{7}$$

We now set $\mathbf{w}_0 = \mathbf{w}_k^*$, where $\mathbf{w}_k^*$ is a local minimizer of $\mathcal{L}_k$. Since $\mathbf{g}^{(k)}(\mathbf{w}_k^*) = \mathbf{0}$, the increment simplifies to

$$\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w}) \approx a_k + \mathbf{g}^{(k+1)}(\mathbf{w}_k^*)^\top(\mathbf{w} - \mathbf{w}_k^*) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_k^*)^\top \mathbf{A}_k(\mathbf{w} - \mathbf{w}_k^*), \tag{8}$$

where $a_k = \mathcal{L}_{k+1}(\mathbf{w}_k^*) - \mathcal{L}_k(\mathbf{w}_k^*)$, $\mathbf{A}_k = \mathbf{H}^{(k+1)}(\mathbf{w}_k^*) - \mathbf{H}^{(k)}(\mathbf{w}_k^*)$.

Subspace reduction. Restricting the probe to a D-dimensional subspace turns the increment into a low-dimensional quadratic form. In particular, for the principal curvature subspace $\mathcal{S}_D = \mathrm{Im}(\mathbf{U}_D)$ with parameterization $\mathbf{w} = \mathbf{w}_k^* + \mathbf{U}_D\mathbf{z}$, the criterion depends only on the compressed quantities

$$\mathbf{c}_k = \mathbf{U}_D^\top \mathbf{g}^{(k+1)}(\mathbf{w}_k^*), \qquad \mathbf{B}_k = \mathbf{U}_D^\top \mathbf{A}_k \mathbf{U}_D.$$

Lemma 1 (Reduction on the principal curvature subspace) *Suppose* $\mathrm{supp}(q) \subset (\mathbf{w}_k^* + \mathcal{S}_D) \cap \mathcal{U}_R(\mathbf{w}_k^*)$*. Under the parameterization* $\mathbf{w} = \mathbf{w}_k^* + \mathbf{U}_D\mathbf{z}$*, the induced probing law* $\tilde{q}$ *on* $\mathbb{R}^D$ *satisfies*

$$\Delta_2^{(D)}(k+1) = \int_{\mathbb{R}^D} \left(a_k + \mathbf{c}_k^\top \mathbf{z} + \frac{1}{2}\mathbf{z}^\top \mathbf{B}_k \mathbf{z} \right)^2 \tilde{q}(\mathbf{z})\, d\mathbf{z}. \tag{9}$$

The proof is given in Appendix A.

Main rate result. The main rate bound requires only local boundedness at the sequential minimizers.

Assumption 1 (Uniform boundedness at sequential minimizers) *There exist constants* $M_\ell, M_{\mathbf{g}}, M_{\mathbf{H}} > 0$*, independent of* k*, such that for all* $k \geq 1$ *and all* $i = 1, \ldots, k+1$*,*

$$|\ell_i(\mathbf{w}_k^*)| \leq M_\ell, \qquad \|\mathbf{g}_{k+1}(\mathbf{w}_k^*)\|_2 \leq M_{\mathbf{g}}, \qquad \|\mathbf{H}_i(\mathbf{w}_k^*)\|_2 \leq M_{\mathbf{H}}.$$

This assumption is used in the proof of Theorem 2; see Appendix A.

Theorem 2 (Subspace mean-squared rate) *Suppose Lemma 1 and Assumption 1 hold, and let* $\tilde{q}(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$*. Then*

$$\Delta_2^{(D)}(k+1) \leq \frac{12M_\ell^2 + 3\sigma^2 M_{\mathbf{g}}^2 + 3\sigma^4(D^2+2D)M_{\mathbf{H}}^2}{(k+1)^2} = \mathcal{O}(k^{-2}). \tag{10}$$

The proof is given in Appendix A. Theorem 2 is a no-rate-loss statement under geometric compression. The criterion is evaluated on a D-dimensional curvature-aligned probe rather than under isotropic perturbations in $\mathbb{R}^N$, yet its mean-squared decay matches that of the full-space criterion.

Spectral interpretation. Under an additional stable-principal-directions regime, the compressed Hessian difference $\mathbf{B}_k$ becomes diagonal in the principal basis, and the criterion admits a closed form in terms of leading eigenvalue increments; see Corollary 4 in Appendix A.

Extremality of the top-D eigenspace. The following proposition formalizes the sense in which the top-D choice is canonical within the eigenspace-aligned family.

Proposition 3 (Extremality of the top-curvature subspace; informal) *Assume* $a_k = 0$*,* $\mathbf{c}_k = \mathbf{0}$*, and that* $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ *and* $\mathbf{H}^{(k+1)}(\mathbf{w}_k^*)$ *share a common eigenbasis with non-negative eigenvalue increments. Then among all* D*-dimensional eigenspace-aligned subspaces, the top-*D *principal curvature subspace maximizes the pure quadratic stabilization signal* $\Delta_{2,I}^{\mathrm{quad}}(k+1)$*.*

A precise statement and proof are given in Appendix A.2.

5 Algorithmic estimation at scale

Lemma 1 and Eq. (9) show that, once the probe is restricted to the principal curvature subspace, the criterion is determined by three compressed objects: the scalar value gap a_k, the projected gradient $\mathbf{c}_k \in \mathbb{R}^D$, and the compressed Hessian difference $\mathbf{B}_k \in \mathbb{R}^{D\times D}$. This leads to a simple estimator taxonomy. One estimator targets the true criterion directly. Two cheaper estimators target its local quadratic surrogate. All three share the same first step: construct the principal curvature subspace at $\mathbf{w}_k^*$.

Cost notation. Let $C_{\mathrm{fwd}}(m)$ denote the cost of one forward evaluation of $\mathcal{L}_m(\mathbf{w})$, $C_{\mathrm{bwd}}(m)$ the cost of one backward pass, and $C_{\mathrm{HVP}}(m)$ the cost of one Hessian–vector product with $\mathbf{H}^{(m)}(\mathbf{w})$. We write S for the Monte Carlo sample count, D for the subspace dimension, and T_{eig} for the number of eigensolver iterations.

Shared step: principal curvature subspace. For each k, we compute the top-D eigenvectors of $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ using deflated power iteration on Hessian–vector products; other HVP-based iterative eigensolvers such as Lanczos or LOBPCG apply identically. These follow the standard scalable second-order toolkit initiated by Pearlmutter and used in modern Hessian-analysis frameworks such as PyHessian [Pearlmutter, 1994, Yao et al., 2020]. If each iteration requires $\mathcal{O}(D)$ Hessian–vector products, the one-time subspace construction cost is $\mathcal{O}\big(T_{\text{eig}}\, D\, C_{\text{HVP}}(k)\big)$. This cost is shared by all subspace-based estimators below.

Direct Monte Carlo for the true criterion. The most faithful estimator samples directly from the subspace probe: $\mathbf{z}_s \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$, $\mathbf{w}_s = \mathbf{w}_k^* + \mathbf{U}_D \mathbf{z}_s$, $s = 1, \ldots, S$. This gives

$$\widehat{\Delta}_{2,\text{dir}}^{(D)}(k+1) = \frac{1}{S} \sum_{s=1}^{S} \Big(\mathcal{L}_{k+1}(\mathbf{w}_s) - \mathcal{L}_k(\mathbf{w}_s) \Big)^2. \tag{11}$$

It estimates the true criterion $\Delta_2^{(D)}$ and does not rely on the quadratic approximation. Its post-subspace cost is $\mathcal{O}\left(S\big(ND + C_{\text{fwd}}(k) + C_{\text{fwd}}(k+1)\big)\right)$, typically dominated by the two forward evaluations per sample.

Quadratic surrogate and its coefficients. Under the local quadratic model from Section 4, the increment restricted to the principal curvature subspace is approximated by $a_k + \mathbf{c}_k^\top \mathbf{z} + \frac{1}{2} \mathbf{z}^\top \mathbf{B}_k \mathbf{z}$, where $a_k = \mathcal{L}_{k+1}(\mathbf{w}_k^*) - \mathcal{L}_k(\mathbf{w}_k^*)$, $\mathbf{c}_k = \mathbf{U}_D^\top \mathbf{g}^{(k+1)}(\mathbf{w}_k^*)$, and $\mathbf{B}_k = \mathbf{U}_D^\top \big(\mathbf{H}^{(k+1)}(\mathbf{w}_k^*) - \mathbf{H}^{(k)}(\mathbf{w}_k^*)\big) \mathbf{U}_D$. Assembling these coefficients requires one additional setup step with cost

$$\mathcal{O}\left(C_{\text{fwd}}(k) + C_{\text{fwd}}(k+1) + C_{\text{bwd}}(k+1) + D(C_{\text{HVP}}(k) + C_{\text{HVP}}(k+1)) + ND^2\right). \tag{12}$$

Quadratic Monte Carlo. Replacing the true increment in (11) by the quadratic surrogate yields

$$\widehat{\Delta}_{2,\text{quadMC}}^{(D)}(k+1) = \frac{1}{S} \sum_{s=1}^{S} \left(a_k + \mathbf{c}_k^\top \mathbf{z}_s + \frac{1}{2} \mathbf{z}_s^\top \mathbf{B}_k \mathbf{z}_s \right)^2, \qquad \mathbf{z}_s \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D). \tag{13}$$

After the coefficient setup in (12), its evaluation cost is $\mathcal{O}(SD^2)$.

Gaussian-moment estimator. Because the surrogate is quadratic in $\mathbf{z}$ and the probe is Gaussian, its expectation can be evaluated in closed form:

$$\widehat{\Delta}_{2,\text{GM}}^{(D)}(k+1) = \mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)} \left[\left(a_k + \mathbf{c}_k^\top \mathbf{z} + \frac{1}{2} \mathbf{z}^\top \mathbf{B}_k \mathbf{z} \right)^2 \right]. \tag{14}$$

Using Gaussian moment identities yields

$$\widehat{\Delta}_{2,\text{GM}}^{(D)}(k+1) = a_k^2 + a_k \sigma^2 \operatorname{Tr}(\mathbf{B}_k) + \sigma^2 \|\mathbf{c}_k\|_2^2 + \frac{\sigma^4}{4} \left(2 \operatorname{Tr}(\mathbf{B}_k^2) + \operatorname{Tr}(\mathbf{B}_k)^2 \right). \tag{15}$$

After setup, this estimator costs $\mathcal{O}(D^2)$.

Summary. The three estimators differ only in where they sit on the fidelity–efficiency spectrum. Direct subspace Monte Carlo targets the true criterion and is therefore the reference estimator for $\Delta_2^{(D)}$. Quadratic Monte Carlo and the Gaussian-moment estimator are much cheaper after setup, but they target only the local quadratic surrogate. Their empirical comparison therefore tests two things at once: computational savings and the practical validity of the quadratic approximation.

6 Experiments

We evaluate four questions[1]: how the proposed criteria decay with effective sample size, when the subspace criterion preserves the full-space mean-squared signal, for which perturbation scales the

[1] Anonymized code for reproducing all figures and tables is available at `https://anonymous.4open.science/r/curvature-probing-landscape`.

Table 2: Subspace-based estimators for $\Delta_2^{(D)}$. All methods share the one-time subspace construction cost $\mathcal{O}(T_{\text{eig}}\, D\, C_{\text{HVP}}(k))$. The most efficient one is Gaussian-moment (GM) estimator.

Estimator	**Target**	**One-time setup**	**Evaluation cost**
Direct MC	true $\Delta_2^{(D)}$	none	$\mathcal{O}\left(S\big(ND + C_{\text{fwd}}(k) + C_{\text{fwd}}(k+1)\big)\right)$
Quadratic MC	quadratic surrogate	Eq. (12)	$\mathcal{O}(SD^2)$
GM	quadratic surrogate	Eq. (12)	$\mathcal{O}(D^2)$

quadratic proxy is accurate, and how the three estimators trade fidelity against efficiency. Surprisingly, we observe that (i) a curvature-aligned probe occupying less than one part in a million of parameter space ($D/N < 10^{-6}$) already reproduces the full-space mean-squared signal to within numerical noise throughout the validated local regime, and (ii) once the subspace has been constructed, the closed-form Gaussian-moment estimator is roughly $18{,}000\times$ faster than direct Monte Carlo without measurable loss of fidelity.

Setup. All experiments use the `nanochat` depth-6 model (tag `d6`), a 107M-parameter decoder-only transformer with rotary position embeddings, grouped-query attention, ReLU activations, and RMSNorm without learnable parameters, evaluated at training step 3500. We use this model because it is small enough to make repeated full-model Hessian–vector products tractable in float32 precision while still retaining nontrivial second-order structure. Autocast is disabled throughout, and we use the SDPA math kernel for numerical stability. Throughout this section, k denotes the number of training sequences defining $\mathcal{L}_k$. Unless stated otherwise, we define $\mathcal{L}_k$ and $\mathcal{L}_{k+1}$ from $k = 8$ sequences and construct the principal curvature subspace at $\mathbf{w}_k^*$ by deflated power iteration on Hessian–vector products; other HVP-based iterative eigensolvers such as Lanczos or LOBPCG apply identically.

Criterion decay under sample growth. We first test whether the criteria from Sections 3–4 exhibit the predicted stabilization trend as the effective sample size grows. To do so, we evaluate the pointwise criterion Δ_1, the isotropic mean-squared criterion Δ_2, and the curvature-aware subspace criterion $\Delta_2^{(D)}$ across a range of sample sizes k.

Figure 2 shows that all criteria decrease with k, but at different rates. The pointwise criterion decays more slowly, whereas the mean-squared criteria are orders of magnitude smaller and follow the steeper trend suggested by the quadratic scaling argument. The subspace criteria remain close in scale to the full-space mean-squared criterion while restricting the probe to a much smaller set of directions.

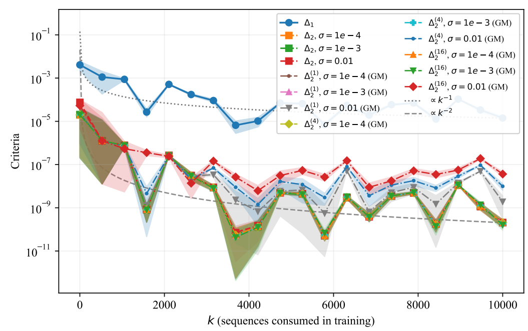

Figure 2: **Decay of stabilization criteria with sample size.** Comparison of Δ_1, Δ_2, and $\Delta_2^{(D)}$ as functions of k.

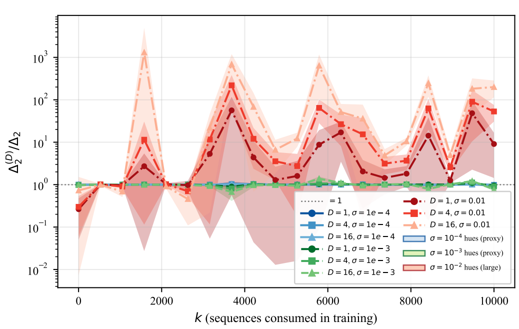

Figure 3: **Subspace criterion relative to the full-space criterion.** Ratio $\Delta_2^{(D)}/\Delta_2$ across sample size k, for several dimensions D and scales σ.

Subspace versus full-space criterion. We next test whether the curvature-aligned subspace criterion preserves the full-space mean-squared signal. For several values of D and σ, we track the ratio $\Delta_2^{(D)}/\Delta_2$ as a function of k.

Figure 3 reveals a clear regime split. For $\sigma = 10^{-4}$ and $\sigma = 10^{-3}$, the ratio stays close to 1 across the full range of k, indicating that curvature-aligned probing preserves essentially the same stabilization signal as the full-space criterion. For $\sigma = 10^{-2}$, the ratio departs strongly from 1, becomes much more variable, and depends clearly on subspace dimension,

Table 3: Wall-clock times (seconds; mean $\pm$ sample standard deviation over five seeds) for the three estimators of $\Delta_2^{(D)}$ on the nanochat `d6` model ($D = 45$, step 3500). Stage I: top-D Hessian eigenvectors (shared). Stage II: gradient and compressed Hessian assembly (proxy estimators only). Stage III: criterion evaluation.

Estimator	Stage I (s)	Stage II (s)	Stage III (s)
Direct MC	17.53 ± 0.53	—	2.197 ± 0.013
Quadratic MC	17.53 ± 0.53	1.777 ± 0.014	$(3.68 \pm 0.05) \times 10^{-3}$
GM	17.53 ± 0.53	1.777 ± 0.014	$(1.22 \pm 0.03) \times 10^{-4}$

with larger D producing systematically larger values. This behavior is consistent with leaving the local regime in which subspace compression remains faithful to the full-space observable.

Quadratic proxy validity. The theory and the proxy estimators of Section 5 rely on the local quadratic model in Eqs. (6)–(8). Before using these estimators, we therefore identify the perturbation scales for which the approximation is accurate. For a fixed checkpoint $\mathbf{w}_k^*$, we draw isotropic Gaussian perturbations $\boldsymbol{\delta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ and compare the true local loss increment $\mathcal{L}_k(\mathbf{w}_k^* + \boldsymbol{\delta}) - \mathcal{L}_k(\mathbf{w}_k^*)$ to its second-order Taylor approximation $\mathbf{g}^{(k)\top}\boldsymbol{\delta} + \frac{1}{2}\boldsymbol{\delta}^\top \mathbf{H}^{(k)} \boldsymbol{\delta}$.

Figure 4 shows a clear local regime: the approximation error remains low and nearly flat up to about $\sigma \approx 10^{-3}$, and then rises sharply for larger perturbations. We therefore restrict the proxy-based estimators to $\sigma \leq 10^{-3}$ in the remaining experiments.

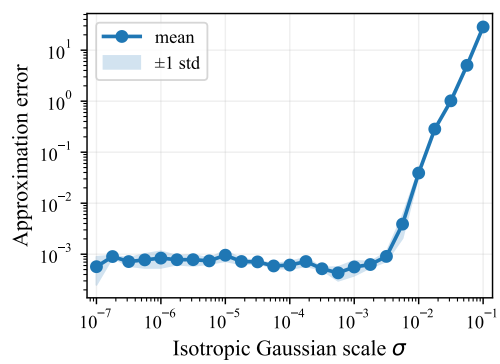

Figure 4: Relative error of the quadratic Taylor approximation versus perturbation scale σ (mean and standard deviation across seeds, nanochat `d6`, step 3500).

Estimator fidelity and computational trade-offs. We finally compare the three estimators from Section 5: direct subspace Monte Carlo, quadratic Monte Carlo, and the Gaussian-moment (GM) estimator. Direct MC targets the true criterion $\Delta_2^{(D)}$, whereas Quadratic MC and GM target its local quadratic surrogate.

Figure 5 shows that both Monte Carlo estimators converge toward the GM value as S increases. Figure 6 shows that the discrepancy between Direct MC and GM stays small throughout the validated local regime, but grows at the largest tested perturbation scale.

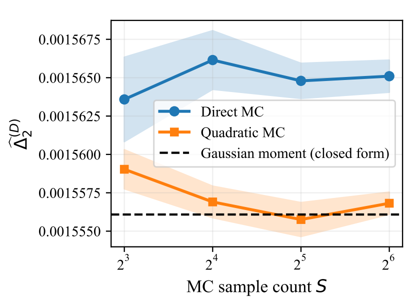

Figure 5: Convergence of Direct MC and Quadratic MC estimates of $\widehat{\Delta}_2^{(D)}$ to the Gaussian-moment closed form as the sample count S increases, at $D = 10$ and $\sigma = 10^{-3}$.

Table 3 shows that the shared subspace-construction stage dominates the total runtime. Once that cost has been paid, GM is by far the cheapest estimator: its evaluation stage is about $18{,}000\times$ faster than Direct MC. Taken together, these results suggest a simple practical picture: Direct MC is the reference estimator for the true criterion, but in the validated local regime the GM proxy provides nearly the same signal at negligible additional cost.

7 Limitations

Our analysis is inherently local. The main theoretical result is derived under a second-order approximation of the increment field near a trained solution $\mathbf{w}_k^*$, so its interpretation is restricted to perturbation regimes in which that approximation is accurate. Accordingly, the proxy estimators from Section 5 should be viewed as local surrogates rather than globally faithful approximations. Theorem 2 is a one-sided rate statement: it certifies that geometric compression preserves the $\mathcal{O}(k^{-2})$ decay, but we do not claim tightness of the $(k+1)^{-2}$ bound or of its $\sigma^4(D^2+2D)$ dependence, and the constant carries a D^2 term that trades against the ambient-dimension factor only for moderate D.

The top-D principal-curvature subspace is motivated by empirical anisotropy and supported by Proposition 3, but this extremality result holds only in an idealized simultaneously-diagonalizable quadratic regime. More general non-quadratic or rapidly drifting regimes may favor adaptive subspaces, and Assumption 1 is itself a local statement whose plausibility near sequential minimizers we do not prove beyond boundedness.

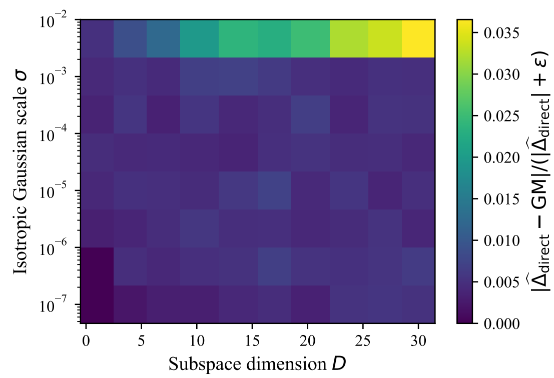

Figure 6: Relative error $|\widehat{\Delta}_{\mathrm{direct}} - \mathrm{GM}|/(|\widehat{\Delta}_{\mathrm{direct}}| + \varepsilon)$ over subspace dimension D and perturbation scale σ.

Our empirical study is intentionally narrow. We focus on a 107M-parameter decoder-only transformer because repeated full-model second-order computations are still feasible in float32. Additional ablations on other architectures and sizes showed a similar qualitative picture, but are omitted for brevity. This scope does not guarantee that the same geometric effects or computational trade-offs transfer unchanged to much larger models or different training regimes. Finally, although Hessian–vector products and iterative eigensolvers are far cheaper than explicit Hessian construction, they still add nontrivial overhead, with subspace construction dominating the cost in our setting.

8 Discussion

Our main claim is conceptual as much as technical: under sample growth, stabilization depends not only on which functional of the increment field is aggregated, but also on how the field is probed. The proposed criterion is therefore not just a lower-dimensional version of an existing mean-squared observable; it makes the probing law explicit and aligns it with the anisotropic local geometry of the empirical landscape.

Within the local quadratic regime, this geometric restriction preserves the canonical $\mathcal{O}(k^{-2})$ mean-squared decay while replacing ambient-dimension curvature dependence by dependence on the probing dimension D. Empirically, the subspace criterion tracks the full-space mean-squared signal throughout the validated local regime, and the Gaussian-moment proxy reproduces the same signal much faster than direct Monte Carlo once the subspace is available.

More broadly, curvature-aligned probing is one instance of a larger family of geometry-aware local observables. If stabilization is viewed as an observational problem, other structured probing laws may reveal aspects of landscape deformation under data growth that curvature alone does not capture. For example, data-dependent subspaces spanned by gradients of influential examples fit the same framework and suggest a broader class of geometry- and data-aware observables compatible with the estimator machinery developed here.

9 Conclusion

We introduced a unified view of local loss-landscape stabilization and proposed a curvature-aligned subspace criterion based on the top-D Hessian eigenspace near a trained solution. Under a local quadratic model, this criterion preserves the full-space mean-squared $\mathcal{O}(k^{-2})$ rate with dependence on the probing dimension D, and admits scalable estimators that are efficient in the valid local regime. This enables a concrete quantitative question—how much of one-sample landscape deformation is concentrated in the dominant curvature modes—to be answered with a closed-form observable. We do not claim that the top-D eigenspace is universally optimal beyond the eigenspace-aligned quadratic regime, nor that a single $\mathcal{O}(k^{-2})$ bound settles the sample-growth behavior of modern training pipelines; both are natural directions for future work.

References

Hao Li, Zheng Xu, Gavin Taylor, Christoph Studer, and Tom Goldstein. Visualizing the loss landscape of neural nets. In *Advances in Neural Information Processing Systems*, 2018a. URL `https://proceedings.neurips.cc/paper/2018/hash/a41b3bb3e6b050b6c9067c67f663b915-Abstract.html`.

Levent Sagun, Utku Evci, Vincent Ugur Guney, Yann Dauphin, and Léon Bottou. Empirical analysis of the hessian of over-parametrized neural networks, 2017.

Behrooz Ghorbani, Shankar Krishnan, and Ying Xiao. An investigation into neural net optimization via hessian analysis, 2019.

Vardan Papyan. The full spectrum of deepnet hessians at scale: Dynamics with SGD training and sample size, 2019.

Yichu Xu, Xin-Chun Li, Lan Li, and De-Chuan Zhan. Visualizing, rethinking, and mining the loss landscape of deep neural networks, 2024.

Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. In *International Conference on Learning Representations*, 2019.

Chunyuan Li, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski. Measuring the intrinsic dimension of objective landscapes. In *International Conference on Learning Representations*, 2018b.

Berfin Simsek, François Ged, Arthur Jacot, Francesco Spadaro, Clement Hongler, Wulfram Gerstner, and Johanni Brea. Geometry of the loss landscape in overparameterized neural networks: Symmetries and invariances. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9722–9732, 2021.

Felix Draxler, Kambis Veschgini, Manfred Salmhofer, and Fred A. Hamprecht. Essentially no barriers in neural network energy landscape. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1309–1318, 2018.

Timur Garipov, Pavel Izmailov, Dmitrii Podoprikhin, Dmitry P. Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of dnns. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Vladimir N. Vapnik. *Statistical Learning Theory*. Wiley, 1998.

Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.

Olivier Bousquet and André Elisseeff. Stability and generalization. *Journal of Machine Learning Research*, 2:499–526, 2002.

Moritz Hardt, Benjamin Recht, and Yoram Singer. Train faster, generalize better: Stability of stochastic gradient descent. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1225–1234. PMLR, 2016. URL `https://proceedings.mlr.press/v48/hardt16.html`.

Olivier Bousquet, Yegor Klochkov, and Nikita Zhivotovskiy. Sharper bounds for uniformly stable algorithms. In *Proceedings of the Thirty Third Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pages 125–135. PMLR, 2020.

Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1885–1894, 2017.

Ryan Giordano, Tamara Broderick, and Michael I. Jordan. A swiss army infinitesimal jackknife. In *Proceedings of Machine Learning Research*, volume 89, pages 1139–1147, 2019.

Pang Wei Koh, Kai-Siang Ang, Hubert H. K. Teo, and Percy Liang. On the accuracy of influence functions for measuring group effects. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Samyadeep Basu, Xuchen You, and Soheil Feizi. On second-order group influence functions for black-box predictions. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2582–2591, 2020.

Vitaly Feldman and Jan Vondrák. High probability generalization bounds for uniformly stable algorithms with applications. In *Proceedings of the Thirty-Second Conference on Learning Theory*, volume 99 of *Proceedings of Machine Learning Research*, pages 1330–1349. PMLR, 2019.

Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations*, 2017.

Laurent Dinh, Razvan Pascanu, Samy Bengio, and Yoshua Bengio. Sharp minima can generalize for deep nets. In *International Conference on Machine Learning*, 2017.

Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations*, 2021.

Yann N. Dauphin, Atish Agarwala, and Hossein Mobahi. Neglected Hessian component explains mysteries in sharpness regularization. In *Advances in Neural Information Processing Systems*, 2024.

Haocheng Luo, Tuan Truong, Tung Pham, Mehrtash Harandi, Dinh Phung, and Trung Le. Explicit eigenvalue regularization improves sharpness-aware minimization. In *Advances in Neural Information Processing Systems*, 2024.

Sidak Pal Singh, Gregor Bachmann, and Thomas Hofmann. Analytic insights into structure and rank of neural network hessian maps. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

Yikai Wu, Xingyu Zhu, Chenwei Wu, Annie Wang, and Rong Ge. Dissecting hessian: Understanding common structure of hessian in neural networks. *arXiv preprint arXiv:2010.04261*, 2020.

Barak A. Pearlmutter. Fast exact multiplication by the Hessian. *Neural Computation*, 6(1):147–160, 1994.

Zhewei Yao, Amir Gholami, Kurt Keutzer, and Michael W. Mahoney. PyHessian: Neural networks through the lens of the Hessian. In *2020 IEEE International Conference on Big Data*, pages 581–590, 2020. doi: 10.1109/BigData50022.2020.9378171.

A Proofs

A.1 Spectral interpretation under stable principal directions

The rate bound in Theorem 2 does not require any alignment between the eigenspaces of $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ and $\mathbf{H}^{(k+1)}(\mathbf{w}_k^*)$. For interpretation, it is useful to isolate a more structured regime in which the leading curvature directions remain stable across the one-sample increment.

Assumption 2 (Stable principal directions) *The eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_D$ associated with the D largest eigenvalues of $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ are also eigenvectors of $\mathbf{H}^{(k+1)}(\mathbf{w}_k^*)$:*

$$\mathbf{H}^{(k+1)}(\mathbf{w}_k^*)\,\mathbf{u}_i = \lambda_i^{(k+1)}\mathbf{u}_i, \qquad i = 1, \dots, D.$$

Under Assumption 2,

$$\mathbf{B}_k = \operatorname{diag}\bigl(\lambda_1^{(k+1)} - \lambda_1^{(k)}, \dots, \lambda_D^{(k+1)} - \lambda_D^{(k)}\bigr),$$

so the compressed Hessian difference becomes diagonal in the principal basis.

Corollary 4 (Spectral closed form under vanishing value and linear terms) *Suppose Assumption 2 holds, $\tilde{q}(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_D)$, and $a_k = 0$, $\mathbf{c}_k = \mathbf{0}$. Then*

$$\Delta_2^{(D)}(k+1) = \frac{\sigma^4}{4}\left(2\sum_{i=1}^{D}(\lambda_i^{(k+1)} - \lambda_i^{(k)})^2 + \left(\sum_{i=1}^{D}(\lambda_i^{(k+1)} - \lambda_i^{(k)})\right)^2\right). \tag{16}$$

Corollary 4 isolates the pure quadratic regime. The assumptions $a_k = 0$ and $\mathbf{c}_k = \mathbf{0}$ are idealizations, but they become increasingly natural when the value gap is small and $\mathbf{w}_k^*$ lies close to a minimizer of $\mathcal{L}_{k+1}$ as well. In that regime, the criterion becomes a direct function of the leading eigenvalue increments.

A.2 Lemma: Extremality of the top-curvature subspace

Lemma 5 (Extremality of the top-curvature subspace) *Assume that $a_k = 0$, $\mathbf{c}_k = \mathbf{0}$. Suppose that $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ and $\mathbf{H}^{(k+1)}(\mathbf{w}_k^*)$ are simultaneously diagonalizable with a common orthonormal eigenbasis $\{\mathbf{u}_i\}_{i=1}^N$, and let $\delta_i = \lambda_i^{(k+1)} - \lambda_i^{(k)}$, $i = 1, \ldots, N$. Assume further that $\delta_1 \geq \delta_2 \geq \cdots \geq \delta_N \geq 0$. For any index set $I \subset \{1, \ldots, N\}$ with $|I| = D$, let $\mathcal{S}_I = \operatorname{span}\{\mathbf{u}_i : i \in I\}$, and define the corresponding quadratic subspace criterion by restricting (4) to $\mathbf{w}_k^* + \mathcal{S}_I$ under the local quadratic model, with Gaussian coordinates $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_D)$. Then*

$$\Delta_{2,I}^{\mathrm{quad}}(k+1) = \frac{\sigma^4}{4}\left(2\sum_{i\in I}\delta_i^2 + \left(\sum_{i\in I}\delta_i\right)^2\right), \tag{17}$$

and the maximum over all index sets I with $|I| = D$ is attained at $I^* = \{1, \ldots, D\}$. Equivalently, the top-D eigenspace of $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ maximizes the pure quadratic stabilization signal within the eigenspace-aligned family.

Under the simultaneous diagonalizability assumption, $\mathbf{B}_k = \operatorname{diag}(\delta_1, \ldots, \delta_N)$ restricted to $\mathcal{S}_I$ gives $\mathbf{B}_{k,I} = \operatorname{diag}(\delta_i : i \in I)$. With $a_k = 0$, $\mathbf{c}_k = \mathbf{0}$, Lemma 1 and the Gaussian moment identities of Theorem 2 yield (17). The function

$$f(I) = 2\sum_{i\in I}\delta_i^2 + \left(\sum_{i\in I}\delta_i\right)^2$$

is maximized by choosing the D largest δ_i values, i.e., $I^* = \{1, \ldots, D\}$, since all terms are non-negative and $\delta_1 \geq \cdots \geq \delta_N \geq 0$.

A.3 Proof of Lemma 1

Since the columns of $\mathbf{U}_D \in \mathbb{R}^{N\times D}$ are orthonormal, every point of $\mathbf{w}_k^* + \mathcal{S}_D$ has a unique representation

$$\mathbf{w} = \mathbf{w}_k^* + \mathbf{U}_D\mathbf{z}, \qquad \mathbf{z} \in \mathbb{R}^D.$$

Substituting into the local quadratic model (8) and using $\mathbf{g}^{(k)}(\mathbf{w}_k^*) = \mathbf{0}$ gives

$$\begin{aligned}\mathcal{L}_{k+1}(\mathbf{w}) - \mathcal{L}_k(\mathbf{w}) &= a_k + \mathbf{g}^{(k+1)}(\mathbf{w}_k^*)^\top\mathbf{U}_D\mathbf{z} + \frac{1}{2}(\mathbf{U}_D\mathbf{z})^\top\mathbf{A}_k(\mathbf{U}_D\mathbf{z})\\ &= a_k + \mathbf{c}_k^\top\mathbf{z} + \frac{1}{2}\mathbf{z}^\top\mathbf{B}_k\mathbf{z}.\end{aligned}$$

Under the parameterization $\mathbf{w} = \mathbf{w}_k^* + \mathbf{U}_D\mathbf{z}$, the density q supported on $\mathbf{w}_k^* + \mathcal{S}_D$ induces a density $\tilde{q}$ on $\mathbb{R}^D$. Integrating yields (9).

A.4 Proof of Theorem 2

By Lemma 1,

$$\Delta_2^{(D)}(k+1) = \mathbb{E}_{\tilde{q}(\mathbf{z})}\left[\left(a_k + \mathbf{c}_k^\top\mathbf{z} + \frac{1}{2}\mathbf{z}^\top\mathbf{B}_k\mathbf{z}\right)^2\right], \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_D).$$

For any $x, y, z \in \mathbb{R}$,

$$(x+y+z)^2 \le 3(x^2+y^2+z^2),$$

hence

$$\Delta_2^{(D)}(k+1) \le 3a_k^2 + 3\,\mathbb{E}\big[(\mathbf{c}_k^\top \mathbf{z})^2\big] + \frac{3}{4}\,\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_k \mathbf{z})^2\big]. \tag{18}$$

Zero-order term. Using (5) at $\mathbf{w}_k^*$,

$$a_k = \mathcal{L}_{k+1}(\mathbf{w}_k^*) - \mathcal{L}_k(\mathbf{w}_k^*) = \frac{1}{k+1}\Big(\ell_{k+1}(\mathbf{w}_k^*) - \mathcal{L}_k(\mathbf{w}_k^*)\Big).$$

Therefore

$$|a_k| \le \frac{1}{k+1}\Big(|\ell_{k+1}(\mathbf{w}_k^*)| + |\mathcal{L}_k(\mathbf{w}_k^*)|\Big).$$

Moreover,

$$|\mathcal{L}_k(\mathbf{w}_k^*)| = \left|\frac{1}{k}\sum_{i=1}^{k} \ell_i(\mathbf{w}_k^*)\right| \le \frac{1}{k}\sum_{i=1}^{k} |\ell_i(\mathbf{w}_k^*)| \le M_\ell,$$

and also $|\ell_{k+1}(\mathbf{w}_k^*)| \le M_\ell$. Hence

$$|a_k| \le \frac{2M_\ell}{k+1}, \qquad a_k^2 \le \frac{4M_\ell^2}{(k+1)^2}. \tag{19}$$

Linear term. Since $\mathbf{g}^{(k)}(\mathbf{w}_k^*) = \mathbf{0}$,

$$\mathbf{b}_k = \mathbf{g}^{(k+1)}(\mathbf{w}_k^*) = \frac{1}{k+1}\mathbf{g}_{k+1}(\mathbf{w}_k^*), \qquad \mathbf{c}_k = \mathbf{U}_D^\top \mathbf{b}_k.$$

Because $\mathbf{U}_D$ has orthonormal columns,

$$\|\mathbf{c}_k\|_2 \le \|\mathbf{U}_D^\top\|_2\,\|\mathbf{b}_k\|_2 \le \frac{M_{\mathbf{g}}}{k+1}.$$

For $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$,

$$\mathbb{E}\big[(\mathbf{c}_k^\top \mathbf{z})^2\big] = \mathbf{c}_k^\top \mathbb{E}[\mathbf{z}\mathbf{z}^\top]\mathbf{c}_k = \sigma^2 \|\mathbf{c}_k\|_2^2 \le \frac{\sigma^2 M_{\mathbf{g}}^2}{(k+1)^2}. \tag{20}$$

Quadratic term. The matrix $\mathbf{B}_k$ is symmetric. For $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$,

$$\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_k \mathbf{z})^2\big] = 2\sigma^4 \operatorname{Tr}(\mathbf{B}_k^2) + \sigma^4 \operatorname{Tr}(\mathbf{B}_k)^2. \tag{21}$$

For symmetric $\mathbf{B}_k$,

$$\operatorname{Tr}(\mathbf{B}_k^2) \le D\|\mathbf{B}_k\|_2^2, \qquad |\operatorname{Tr}(\mathbf{B}_k)| \le D\|\mathbf{B}_k\|_2,$$

so

$$\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_k \mathbf{z})^2\big] \le \sigma^4 (D^2 + 2D)\|\mathbf{B}_k\|_2^2. \tag{22}$$

Next,

$$\|\mathbf{B}_k\|_2 = \|\mathbf{U}_D^\top \mathbf{A}_k \mathbf{U}_D\|_2 \le \|\mathbf{U}_D^\top\|_2\,\|\mathbf{A}_k\|_2\,\|\mathbf{U}_D\|_2 \le \|\mathbf{A}_k\|_2.$$

Using

$$\mathbf{H}^{(m)}(\mathbf{w}_k^*) = \frac{1}{m}\sum_{i=1}^{m} \mathbf{H}_i(\mathbf{w}_k^*),$$

we obtain

$$\begin{aligned}
\mathbf{A}_k &= \mathbf{H}^{(k+1)}(\mathbf{w}_k^*) - \mathbf{H}^{(k)}(\mathbf{w}_k^*) \\
&= \frac{1}{k+1}\sum_{i=1}^{k+1} \mathbf{H}_i(\mathbf{w}_k^*) - \frac{1}{k}\sum_{i=1}^{k} \mathbf{H}_i(\mathbf{w}_k^*) \\
&= \frac{1}{k+1}\Big(\mathbf{H}_{k+1}(\mathbf{w}_k^*) - \mathbf{H}^{(k)}(\mathbf{w}_k^*)\Big).
\end{aligned}$$

Hence

$$\|\mathbf{A}_k\|_2 \le \frac{1}{k+1}\Big(\|\mathbf{H}_{k+1}(\mathbf{w}_k^*)\|_2 + \|\mathbf{H}^{(k)}(\mathbf{w}_k^*)\|_2\Big) \le \frac{2M_{\mathbf{H}}}{k+1},$$

and therefore

$$\|\mathbf{B}_k\|_2 \le \frac{2M_{\mathbf{H}}}{k+1}, \qquad \mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_k \mathbf{z})^2\big] \le \frac{4\sigma^4 (D^2 + 2D) M_{\mathbf{H}}^2}{(k+1)^2}. \tag{23}$$

Conclusion. Substituting (19), (20), and (23) into (18), we obtain

$$\Delta_2^{(D)}(k+1) \le \frac{12M_\ell^2 + 3\sigma^2 M_{\mathbf{g}}^2 + 3\sigma^4(D^2+2D)M_{\mathbf{H}}^2}{(k+1)^2},$$

which is exactly (10).

A.5 Proof of Corollary 4

Under the assumptions $a_k = 0$ and $\mathbf{c}_k = \mathbf{0}$, Lemma 1 gives

$$\Delta_2^{(D)}(k+1) = \frac{1}{4}\,\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_k \mathbf{z})^2\big], \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D).$$

Using (21),

$$\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_k \mathbf{z})^2\big] = 2\sigma^4\,\mathrm{Tr}(\mathbf{B}_k^2) + \sigma^4\,\mathrm{Tr}(\mathbf{B}_k)^2.$$

Under Assumption 2,

$$\mathbf{B}_k = \mathrm{diag}\big(\lambda_1^{(k+1)} - \lambda_1^{(k)}, \ldots, \lambda_D^{(k+1)} - \lambda_D^{(k)}\big).$$

Therefore,

$$\mathrm{Tr}(\mathbf{B}_k) = \sum_{i=1}^{D}\big(\lambda_i^{(k+1)} - \lambda_i^{(k)}\big), \qquad \mathrm{Tr}(\mathbf{B}_k^2) = \sum_{i=1}^{D}\big(\lambda_i^{(k+1)} - \lambda_i^{(k)}\big)^2.$$

Substituting these identities into the previous display yields

$$\Delta_2^{(D)}(k+1) = \frac{\sigma^4}{4}\left(2\sum_{i=1}^{D}(\lambda_i^{(k+1)} - \lambda_i^{(k)})^2 + \left(\sum_{i=1}^{D}(\lambda_i^{(k+1)} - \lambda_i^{(k)})\right)^2\right),$$

which is exactly (16).

A.6 Proof of Lemma 5

Fix an index set $I \subset \{1, \ldots, N\}$ with $|I| = D$ and consider the eigenspace-aligned subspace

$$\mathcal{S}_I = \mathrm{span}\{\mathbf{u}_i : \ i \in I\}.$$

Under the assumptions of the lemma, the linear and value terms vanish, and the Hessian difference is diagonal in the common eigenbasis:

$$\mathbf{A}_k = \mathbf{H}^{(k+1)}(\mathbf{w}_k^*) - \mathbf{H}^{(k)}(\mathbf{w}_k^*) = \sum_{i=1}^{N} \delta_i\, \mathbf{u}_i \mathbf{u}_i^\top, \qquad \delta_i = \lambda_i^{(k+1)} - \lambda_i^{(k)}.$$

Let $\mathbf{U}_I$ denote the matrix whose columns are the vectors $\mathbf{u}_i$, $i \in I$. Then the compressed Hessian difference on $\mathcal{S}_I$ is

$$\mathbf{B}_{k,I} = \mathbf{U}_I^\top \mathbf{A}_k \mathbf{U}_I = \mathrm{diag}(\delta_i : \ i \in I).$$

With Gaussian coordinates $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$, the local quadratic reduction (same argument as in Lemma 1, with $\mathbf{U}_D$ replaced by $\mathbf{U}_I$) and $a_k = 0$, $\mathbf{c}_k = \mathbf{0}$ give

$$\Delta_{2,I}^{\mathrm{quad}}(k+1) = \frac{1}{4}\,\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_{k,I} \mathbf{z})^2\big].$$

Applying the Gaussian quadratic-form identity (21),

$$\mathbb{E}\big[(\mathbf{z}^\top \mathbf{B}_{k,I} \mathbf{z})^2\big] = 2\sigma^4\,\mathrm{Tr}(\mathbf{B}_{k,I}^2) + \sigma^4\,\mathrm{Tr}(\mathbf{B}_{k,I})^2.$$

Since $\mathbf{B}_{k,I}$ is diagonal,

$$\mathrm{Tr}(\mathbf{B}_{k,I}^2) = \sum_{i \in I} \delta_i^2, \qquad \mathrm{Tr}(\mathbf{B}_{k,I}) = \sum_{i \in I} \delta_i.$$

Therefore

$$\Delta_{2,I}^{\text{quad}}(k+1) = \frac{\sigma^4}{4}\left(2\sum_{i\in I}\delta_i^2 + \left(\sum_{i\in I}\delta_i\right)^2\right),$$

which proves (17).

It remains to show that this quantity is maximized when $I = \{1, \ldots, D\}$. Define

$$F(I) = 2\sum_{i\in I}\delta_i^2 + \left(\sum_{i\in I}\delta_i\right)^2.$$

Since $\delta_1 \geq \delta_2 \geq \cdots \geq \delta_N \geq 0$, both terms in $F(I)$ are monotone with respect to replacing a smaller selected δ_j by a larger unselected δ_i. More explicitly, suppose $i \notin I$, $j \in I$, and $\delta_i \geq \delta_j$, and define

$$I' = (I \setminus \{j\}) \cup \{i\}.$$

Then

$$\sum_{\ell\in I'}\delta_\ell = \sum_{\ell\in I}\delta_\ell - \delta_j + \delta_i \geq \sum_{\ell\in I}\delta_\ell,$$

and similarly

$$\sum_{\ell\in I'}\delta_\ell^2 = \sum_{\ell\in I}\delta_\ell^2 - \delta_j^2 + \delta_i^2 \geq \sum_{\ell\in I}\delta_\ell^2.$$

Hence

$$F(I') \geq F(I).$$

By repeatedly exchanging smaller selected indices for larger unselected ones, one reaches the set $I^* = \{1, \ldots, D\}$ without decreasing F. Therefore $F(I)$, and hence $\Delta_{2,I}^{\text{quad}}(k+1)$, is maximized at $I^* = \{1, \ldots, D\}$. This proves the lemma.

B Additional experimental details

Model and numerical setup. All main-text experiments use the `nanochat` depth-6 model (tag `d6`) evaluated at training step 3500. The model is a decoder-only transformer with rotary position embeddings, grouped-query attention, ReLU activations, and RMSNorm without learnable parameters. We use this setup because it is small enough to make repeated full-model Hessian–vector products feasible in float32 precision while still retaining nontrivial transformer loss geometry. Autocast is disabled throughout, and the SDPA math kernel is used for numerical stability.

Losses, checkpoints, and local quantities. Unless stated otherwise, the local criteria are evaluated at a checkpoint denoted by $\mathbf{w}_k^*$. The empirical risks $\mathcal{L}_k$ and $\mathcal{L}_{k+1}$ are formed from the corresponding nested sequence subsets used in the experiment under consideration. The local quadratic quantities in Sections 4 and 5 are computed at the same checkpoint:

$$a_k = \mathcal{L}_{k+1}(\mathbf{w}_k^*) - \mathcal{L}_k(\mathbf{w}_k^*), \qquad \mathbf{c}_k = \mathbf{U}_D^\top \mathbf{g}^{(k+1)}(\mathbf{w}_k^*), \qquad \mathbf{B}_k = \mathbf{U}_D^\top\big(\mathbf{H}^{(k+1)} - \mathbf{H}^{(k)}\big)\mathbf{U}_D.$$

Subspace construction. Top-D Hessian eigendirections are computed from $\mathbf{H}^{(k)}(\mathbf{w}_k^*)$ using Hessian–vector products and iterative eigensolvers, as described in Section 5. Unless stated otherwise, the principal curvature subspace is recomputed for each evaluated checkpoint.

Criterion estimation. For Δ_1, we evaluate the one-point increment directly at $\mathbf{w}_k^*$. For Δ_2 and $\Delta_2^{(D)}$, Monte Carlo estimates use Gaussian perturbations with the perturbation scales and sample counts reported in the corresponding figure captions. Direct subspace Monte Carlo samples perturbations in the principal curvature subspace, while Quadratic MC and the Gaussian-moment estimator use the compressed surrogate coefficients $(a_k, \mathbf{c}_k, \mathbf{B}_k)$. Unless stated otherwise, proxy-based estimators are used only in the perturbation regime where the quadratic approximation is empirically validated.

Quadratic proxy validation. For the proxy-validity experiment in the main text, we draw isotropic Gaussian perturbations

$$\boldsymbol{\delta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$$

and compare the true local loss increment

$$\mathcal{L}_k(\mathbf{w}_k^* + \boldsymbol{\delta}) - \mathcal{L}_k(\mathbf{w}_k^*)$$

to its second-order Taylor approximation

$$\mathbf{g}^{(k)\top}\boldsymbol{\delta} + \frac{1}{2}\boldsymbol{\delta}^\top \mathbf{H}^{(k)}\boldsymbol{\delta}.$$

The reported relative errors are averaged over random seeds and perturbation samples.

Decay under sample growth. For the sample-growth experiment in the main text, we evaluate Δ_1, Δ_2, and $\Delta_2^{(D)}$ across a grid of effective sample sizes k. The corresponding curves are intended to test the predicted decay of the stabilization criteria and to compare pointwise, isotropic, and curvature-aligned probes on the same training trajectory. When the experiment uses multiple random seeds, the plotted values are averaged across seeds and displayed with the uncertainty convention specified in the figure caption.

Runtime measurements. Wall-clock times in Table 3 are measured separately for three stages: subspace construction, surrogate-coefficient assembly, and criterion evaluation. The first stage is shared by all estimators. Reported values are averaged over five random seeds and shown as mean $\pm$ sample standard deviation.

Recorded outputs. For each run, we record the estimated criteria, proxy-validation diagnostics, subspace-comparison results, and estimator timing statistics needed to generate the figures and tables in the main text.

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